

Engagement in LLM chatbot-supported learning: The pivotal roles of GenAI competency and emotion

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ABSTRACT

Student engagement, widely recognised as a key determinant of learning outcomes, warrants greater attention to its antecedent mechanisms for effective learning design. As LLM-based chatbots (hereafter chatbot) are increasingly used to support student learning, understanding what drives student engagement in such environments becomes especially important. Guided by appraisal theory, this study proposes a theoretically grounded model of students' engagement in chatbot-supported learning. Unlike prior studies that focus on isolated antecedents, the model integrates learners' capacity appraisals, including technology-related (GenAI competency) and task-related (academic self-efficacy) dimensions, with in-situ cognitive appraisals (perceived usefulness and ease of use of the chatbot) and affective appraisals (emotions experienced when using the chatbot) to explain student engagement. Data were collected after students completed chatbot-supported learning tasks, resulting in 234 responses. Structural equation modelling and mediation analyses revealed that GenAI competency had the strongest relationship with student engagement. Emotion fully mediated the effect of perceived usefulness on engagement, while perceived usefulness partially mediated the effect of perceived ease of use. Notably, academic self-efficacy showed no significant direct association with engagement after accounting for the effects of other variables. These findings denote a differentiated model of student engagement in the context of chatbot-supported learning, highlighting the critical role of GenAI competency and positive emotions. They contribute to the literature on learning engagement and offer practical insights for designing AI-integrated learning environments that promote and sustain meaningful student engagement.

1. Introduction

Learning engagement is a critical research focus due to its demonstrated impact on learning persistence (Jung & Lee, 2018; Tinto, 2022), academic achievement (Lei et al., 2018; Wong et al., 2024), and psychological well-being (Boulton et al., 2019; Pan et al., 2023). With the rapid advancement of Generative Artificial Intelligence (GenAI), Large Language Model-based chatbots (hereafter chatbot) now offer advanced human-like learning capabilities, providing personalised responses and interactions that increasingly resemble authentic human communication (Ivanov et al., 2024; Yuan & Liu, 2025). Consequently, there is growing interest in harnessing chatbot to enhance student learning (Ahuerma, 2024; Gökoğlu & Erdoğan, 2025).

Despite the importance of student engagement, empirical research on engagement chatbot-supported learning remains scarce. Existing

studies have predominantly investigated students' attitudes toward chatbot or their willingness to use them (Hsiao & Tang, 2025; Law, 2024), neither of which can precisely capture students' actual engagement. Moreover, these studies often rely on surveys disconnected from specific learning contexts, limiting their ability to reflect how students genuinely engage in chatbot-supported learning activities. Although a few recent studies have begun to explore antecedents of student engagement using cluster analysis (Hao et al., 2024; Hossain et al., 2025), they do not propose or empirically test a theoretical model of engagement antecedents. This underscores the need to move beyond descriptive or exploratory analyses and to examine student engagement within authentic chatbot-supported learning scenarios through a theoretically grounded model of engagement antecedents. Specifically, in this study, antecedents of engagement refer to individual learners' appraisal-based cognitive and affective processes that precede and shape

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their engagement during chatbot-supported learning, rather than broader contextual or institutional factors.

To address this research gap, this study adopts appraisal theory (Lazarus, 1991a, 1991b), which posits that individuals undergo a sequential appraisal process in which their foundational understanding and judgements about tools influence how they evaluate the performance of the tools. These judgements, together with evaluations of the learning task, constitute cognitive appraisals that shape individuals' emotional responses, which ultimately influence engagement behaviours. By articulating the cognitive-emotional sequence among these factors, appraisal theory provides a robust framework for understanding how students engage in chatbot-supported learning. Building on this theoretical foundation, the present study conceptualises students' GenAI competency as their foundational understanding of GenAI tools, hypothesised to predict their judgements of the perceived ease of use (PE) and perceived usefulness (PU) of specific chatbots in relation to learning tasks. Academic self-efficacy is used to represent students' judgements of their ability to complete learning tasks. This study posits that these cognitive evaluations influence students' learning engagement through their emotional experiences during the learning process. Accordingly, the core research question is: *How do students' cognitive appraisals (GenAI competency, PE, PU, and academic self-efficacy) influence their emotional experiences, and how do these cognitive-emotional processes shape their engagement in chatbot-supported learning?*

This research offers theoretical, methodological, and practical contributions. Theoretically, it extends appraisal theory (Lazarus, 1991a, 1991b) by applying it to the context of chatbot-supported learning engagement and by explicitly incorporating GenAI competency and academic self-efficacy as two main sources of primary appraisal. This recognises that learners' knowledge and skills in leveraging GenAI and their efficacy beliefs in learning tasks are critical for constituting their cognitive appraisals, shaping emotional responses, and ultimately influencing their engagement.

Methodologically, this study collects empirical data embedded within concrete learning tasks, moving beyond general adoption intentions assessed in decontextualised settings to the direct measurement of student engagement processes within authentic chatbot-supported learning activities. This approach makes it possible to uncover the nuanced processes of student engagement in chatbot-supported learning. Practically, the findings offer valuable insights for educators and curriculum designers on how to develop pedagogically meaningful GenAI learning environments, ultimately helping to prepare learners for an educational future increasingly shaped by GenAI technologies (Runge et al., 2025; Wang et al., 2024).

2. Literature review

2.1. Engagement in chatbot-supported learning

Learning engagement refers to the extent of a learner's active involvement, psychological investment, and sustained effort in the learning process (Halverson & Graham, 2019). It goes beyond mere participation or attendance, capturing the extent to which learners exert effort and remain committed to achieving meaningful educational outcomes (NOE et al., 2010; Rajabalee et al., 2020). Prior studies consistently demonstrate that learning engagement significantly influences learners' satisfaction and academic performance (Boulton et al., 2019; Lei et al., 2018). Consistent with the general definition of learning engagement, chatbot-supported learning engagement refers to students' engagement in learning activities supported by chatbot, which can foster student thinking and reflection throughout the learning process (Shao & Chen, 2021; Wang et al., 2024). Learning engagement is highly context-sensitive (Guo et al., 2023; Şeker, 2023), meaning that the antecedent mechanisms driving engagement may differ across learning contexts and tools. As a specific example, chatbot such as ChatGPT and Copilot are increasingly integrated into higher education classrooms to

support student learning (Newmann et al., 2025; Shuhaiber et al., 2025). However, the antecedents of student engagement in such learning environments remain largely unexplored. To address this gap, this study draws on appraisal theory to develop a framework for examining the factors influencing students' engagement in chatbot-supported learning.

2.2. Appraisal theory

Lazarus's appraisal theory (1991a, 1991b) explains that emotional and behavioural responses are shaped through a sequential appraisal process. It begins with primary appraisal, where individuals judge the significance and relevance of a task in relation to their personal goals. This judgement is based on their foundational knowledge and their understanding of both their own capacity and the task. Secondary appraisal follows, involving an evaluation of the resources and coping options available to manage the task. Together, these cognitive appraisals shape emotional experiences, which then influence behaviour. This theoretical framework has been widely adopted in educational research to explore how students' appraisals shape emotional responses and, in turn, engagement behaviours. For instance, Loderer et al. (2020), in their systematic review on emotions in technology-based learning environments, emphasise the central role of cognitive (control and value) appraisals in shaping students' emotional experiences. These emotional experiences, in turn, influence students' cognitive, behavioural, and affective engagement. Their review illustrates how the appraisal-emotion-engagement sequence offers a robust lens for understanding learner behaviour in digital learning contexts. In parallel, appraisal theory has also been applied in technology adoption research, particularly in the context of GenAI. For example, Gursoy and colleagues (Chi et al., 2022; Gursoy et al., 2019; Lin et al., 2020) drew on Lazarus's appraisal theory to examine how individuals' cognitive beliefs about GenAI shape their emotional responses, which subsequently influence behavioural intentions to use AI devices. These findings underscore the applicability of appraisal theory in explaining how individuals interact with AI tools and offer a strong theoretical rationale for using this framework to investigate student engagement in chatbot-supported learning environments.

Building on this theoretical foundation, the present study regards GenAI competency and academic self-efficacy as two primary sources of students' primary appraisal. GenAI competency reflects students' basic knowledge and skills for using GenAI tools. Academic self-efficacy refers to their judgment of their ability to complete learning tasks. Next, GenAI competency further shapes students' secondary appraisal by guiding how they perceive chatbot, particularly in terms of PE and PU when completing learning tasks. These appraisals affect students' emotional experiences during learning, which in turn influence their engagement in learning with chatbot (see Fig. 1 for the conceptual model of this study). In the following section, existing studies will be discussed to substantially support the relationship between the examined variables.

2.3. The impact of GenAI competency on students' chatbot perceptions, emotion, and learning engagement

In this study, GenAI competency refers to students' perceived knowledge, skills, and confidence in using GenAI tools to complete academic and everyday tasks in an effective and appropriate manner (Chiu, 2025; Miao & Shiohira, 2024). Drawing on established technology adoption literature (Davies, 2024; Venkatesh et al., 2012), perceived usefulness (PU) in the present study reflects the extent to which students perceive chatbots as useful for supporting their learning, whereas perceived ease of use (PE) captures students' beliefs about the effort required to employ these tools effectively in learning contexts. A substantial body of technology adoption research has consistently demonstrated the central role of users' digital competency in shaping both PE and PU of technological tools. Individuals with higher levels of digital competency typically possess stronger foundational knowledge

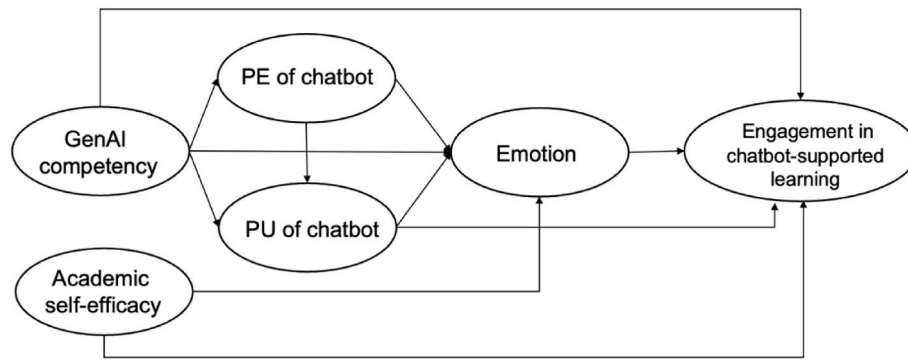


Fig. 1. The conceptual model of this study.

Note: Chatbot means LLM-based chatbots; PE means perceived ease of use, and PU means perceived usefulness.

and operational skills, enabling them to navigate technologies more efficiently when completing tasks. Accumulated mastery experiences in technology use, in turn, increase the likelihood that users perceive a technology as both easy to use and useful for task accomplishment (Antonietti et al., 2022; Nikou & Economides, 2017). This pattern appears to extend to GenAI technologies. Individuals with higher levels of GenAI competency are better equipped with the conceptual understanding and practical skills required to interact effectively with GenAI. As a result, they tend to report higher levels of PE and PU when employing GenAI tools to address tasks and problems (Al-Abdullatif, 2024; Jang, 2024).

GenAI competency also plays an important role in shaping individuals' emotional responses and learning engagement. Recent studies on GenAI competency suggest that individuals with limited GenAI knowledge and skills are more likely to experience negative emotions, such as anxiety, frustration, or disengagement, when interacting with AI tools (Subasi et al., 2024). These emotional responses often arise when individuals encounter difficulties in understanding system outputs or managing interactions with GenAI applications. In contrast, individuals with higher levels of GenAI competency tend to report more positive emotions, including confidence, enjoyment, and trust, as their stronger knowledge base and operational skills enable smoother and more effective interactions with GenAI tools (Zhang et al., 2024).

In addition to emotional responses, existing literature on online learning engagement indicates that learners' technological knowledge and skills are closely associated with their level of engagement in technology-supported learning environments. Prior research has shown that higher levels of technological competency enable learners to make more effective use of digital tools for communication, feedback utilisation, and task completion, which in turn supports sustained learning engagement (Aldowah et al., 2025; Hu & Xiao, 2025). It is therefore reasonable to expect that this positive relationship extends to chatbot-supported learning, where GenAI competency equips learners with the capacity to interact productively with chatbots and remain engaged throughout the learning process. Based on these theoretical considerations and empirical evidence, this study proposes a set of hypotheses concerning the role of GenAI competency in shaping learners' perceptions, emotional responses, and engagement in chatbot-supported learning.

H1. Students' GenAI competency positively predicts their PE of chatbot.

H2. Students' GenAI competency positively predicts their PU of chatbot.

H3. Students' GenAI competency positively predicts their emotions in chatbot-supported learning.

H4. Students' GenAI competency positively predicts their engagement in chatbot-supported learning.

2.4. The impact of academic self-efficacy on emotion and learning engagement

Student academic self-efficacy refers to learners' judgments regarding their capabilities to attain desired learning outcomes (Elias & MacDonald, 2007; Pintrich & De Groot, 1990). As a core component of learning motivation, academic self-efficacy influences how students approach challenges, allocate effort, and persist in the face of academic demands. Academic self-efficacy plays a pivotal role in shaping students' emotional responses in learning. According to Pekrun's Control-Value Theory (2006), self-efficacy serves as a key control appraisal that directly influences academic emotions. When students perceive high control over their academic activities—reflected through strong self-efficacy beliefs—they are more likely to experience positive emotions such as enjoyment, hope, and pride (Hayat et al., 2020). Empirical studies have consistently demonstrated that students who possess high levels of academic self-efficacy experience heightened positive emotions during learning (Hayat et al., 2020; Putwain et al., 2013; Usán Supervía & Quílez Robres, 2021).

A robust positive relationship between students' academic self-efficacy and their learning engagement across diverse educational contexts has been substantially supported in prior studies (Honicke & Broadbent, 2016). Academic self-efficacy functions as a proximal determinant of learners' behavioural investment, influencing the degree to which students engage cognitively and behaviourally in academic tasks (Luo et al., 2023). Students with strong self-efficacy are more likely to initiate and sustain active learning behaviours, demonstrating higher levels of persistence and strategic effort (Wu et al., 2020). In technology-enhanced learning environments, students with high academic self-efficacy are more inclined to put more effort into their studying, with higher levels of confidence and optimism exhibited when achieving learning objectives (Getenet et al., 2024; Huang & Wang, 2023). Based on these theoretical perspectives and empirical findings, this study proposes the following hypotheses:

H5. Student academic self-efficacy positively predicts their emotions in chatbot-supported learning.

H6. Student academic self-efficacy positively predicts their engagement in chatbot-supported learning.

2.5. Linking PE and PU with emotion and learning engagement

Extensive research in educational technology has demonstrated how PE influences both learners' PU and their emotional responses. When individuals perceive a technology as easy to operate, they are more likely to evaluate it as useful, as user-friendly tools reduce the cognitive effort required to achieve task goals, thereby enhancing perceptions of effectiveness and efficiency (Al-Abdullatif, 2024; Nikou & Economides, 2017; Wu & Chen, 2017). Beyond shaping cognitive evaluations, ease of

use is also closely linked to emotional outcomes. Technologies that are intuitive and straightforward to navigate tend to mitigate negative affective states such as anxiety and frustration, while fostering positive experiences like confidence and satisfaction (Huang, 2021, pp. 29–35; Talha et al., 2025).

PU has likewise been shown to elicit positive emotional responses. Studies indicate that users who recognise a technology's usefulness in achieving their goals are more likely to experience satisfaction, enjoyment, and trust (Al-Abdullatif, 2024; Ma et al., 2024; Wang & Yin, 2025). In educational settings, when students see GenAI tools as effectively supporting their learning goals, their emotional resistance usually decreases, and they report greater enjoy interacting with AI-powered systems (Lee et al., 2023; Lu et al., 2019).

Beyond PU's role in shaping emotional responses, it has also been consistently linked to learners' engagement with technology. When students perceive a technology as useful for accomplishing their tasks, they are more willing to invest effort in adopting it for learning (Teo, 2011; Venkatesh et al., 2012). Within the GenAI literature, students who regard GenAI tools as beneficial for achieving academic goals are more inclined to engage actively, sustain attention, and demonstrate greater perseverance in learning tasks (Al-Azawei et al., 2023; Hew et al., 2023). Therefore, the following hypotheses are proposed.

- H7.** Students' PE of chatbot positively predicts their PU of chatbot.
- H8.** Students' PE of chatbot positively predicts their positive emotions in chatbot-supported learning.
- H9.** Students' PU of LLM chatbots positively predicts their positive emotions in chatbot-supported learning.
- H10.** Students' PU positively predicts their engagement in chatbot-supported learning.

2.6. The impact of emotion on learning engagement

Studies across both learning engagement and technology adoption domains have underscored the critical role of emotional responses to learning engagement. In educational psychology, positive emotional states—such as enjoyment, satisfaction, and trust—have consistently been shown to foster deeper cognitive involvement and sustained behavioural engagement (Allcoat & von Mühlenen, 2018; Wang et al., 2022). In the technology adoption literature, when students experience favourable emotions toward GenAI tools, they are more likely to engage meaningfully with the technology, demonstrating persistence, curiosity, and a willingness to explore its full educational potential (D'Errico et al., 2016; Venn et al., 2023). As such, the following hypothesis is proposed:

- H11.** Students' positive emotions positively predicts their engagement in chatbot-supported learning.

3. Methods

3.1. Participants

This study employed a task-embedded survey design to capture students' cognitive appraisals, emotional experiences, and engagement in chatbot-supported learning activities. Participants were master's students from diverse academic backgrounds, including English, business English, physical education, early education, and special education, reflecting the broad disciplinary composition of the cohort. Participants were enrolled in the course (approximately 90 students). Participants ranged in age from 18 to 45 years, with a mean age of 24.3 ($SD = 3.47$). Participants were invited to complete questionnaires following chatbot-supported learning tasks, yielding 234 responses.

3.2. Procedures

The study was conducted in a master-level education course on curriculum design at a university in Hong Kong. The first author was the lecturer of this course. Data were collected across multiple curriculum design activities embedded in the course. Each task was related to a core component of curriculum design, such as the design of learning objectives, learning activities, and assessment activities. For each task, the lecturer first explained the relevant principles and demonstrated them using authentic examples. Students were then required to complete a 20-min learning task by designing relevant objectives, activities, or assessments in their own subject areas, with support from LLM-based chatbots. Throughout the learning activities, the lecturer remained in class to guide and support students while they worked on the tasks. During the learning process, students were encouraged to use ChatGPT 4.0 or DeepSeek-R2 to support task completion. Specifically, the chatbot could be used to clarify difficult terms, brainstorm ideas, transform initial ideas into concrete lesson plan designs, and evaluate the alignment between students' designs and the principles introduced by the lecturer. This process enabled students to deepen their understanding of curriculum design principles and apply them to lesson planning across different disciplines.

To capture students' learning experiences in relation to each task, surveys were administered after students completed the chatbot-supported learning tasks using Wenjuanxing, a widely used online survey platform in China. Ethical approval was obtained from the first author's university's Research Ethics Committee prior to data collection. Participation was voluntary and anonymous. Students were informed that they could opt out of using the chatbot during learning, decline to disclose their perceptions, or withdraw from the study at any time without academic or personal consequences. Students' names or other identifying information were not collected, ensuring confidentiality and avoiding potential conflicts of interest. Because participation in the survey was voluntary, some questionnaires were not obtained when students were absent from or chose not to complete the survey. These instances constitute unit nonresponse and were therefore not included in the analytic dataset. As a result, a total of 234 responses were retained for the final analysis.

3.3. Measures

GenAI competency. This construct was assessed using a five-item scale adapted from Celik (2023), which measured participants' general knowledge and skills in using GenAI-based tools. A sample item includes: 'I have sufficient knowledge to use GenAI tools'. Responses were recorded on a 7-point Likert scale, ranging from 1 (strongly disagree) to 7 (strongly agree).

Academic self-efficacy. This construct was assessed using four items from a subscale of the Motivated for Learning Strategies Questionnaire (Pintrich & De Groot, 1990). An example item is: 'Regarding today's task, I am sure I can do an excellent job on the problems and tasks assigned'. Responses were rated on a 7-point Likert scale from strongly disagree to strongly agree.

PE. The three-item scale for PE, developed by Davis (1989), assessed how easily student teachers could interact with the chatbot. A sample item includes: 'Regarding today's task, my interaction with chatbots is clear and understandable'. Responses were provided on a 7-point Likert scale, ranging from extremely unlikely to extremely likely.

PU. PU was measured using a four-item scale originally developed by Davis (1989), which evaluated participants' perception of the chatbot's usefulness in completing learning tasks. An example item was as follows: 'Regarding today's task, I think the chatbots make it easier to do my task'. Participants were asked to answer each item on a 7-point Likert scale ranging from extremely unlikely to extremely likely.

Emotion. Student emotion experience was captured by a five-item bipolar scale adapted from Lu et al. (2019). Each item presented a

semantic differential between two opposing emotional states. A sample item was as follows: ‘When I use chatbots to finish today’s task, I feel Bored – Relaxed’.

Engagement in chatbot-supported learning. Learning engagement was assessed using a six-item scale developed by Wang et al. (2024). Participants rated their level of engagement in chatbot-supported learning experiences on a 6-point Likert scale, ranging from “strongly disagree” to “strongly agree”. A sample item includes: ‘Regarding today’s task, I am involved in the learning experiences using chatbot’.

3.4. Data analysis

A structured, multi-step approach was employed to ensure the robustness and validity of the findings. After deleting outliers and erroneous values, the normality of the data was examined to assess whether skewness and kurtosis for each of the 27 items fell within acceptable ranges. Following this, the internal consistency of the questionnaire was computed by computing Cronbach’s alpha (α) for each construct. All six constructs should exhibit Cronbach’s α values above 0.8, exceeding the threshold recommended by Kline (2023) and confirming the reliability of the scales.

Since the study relied on self-reported data, we assessed the potential impact of common method variance (CMV) through Harman’s single-factor test (Podsakoff et al., 2003), ensuring that no single factor accounted for the majority of variance, thereby minimising concerns of measurement bias.

Next, we conducted confirmatory factor analysis (CFA) to validate the factorial structure of each scale. According to Stevens (2009), factor loadings should be above 0.50 to establish construct validity.

Following the validation of measurement properties, we calculated the descriptive statistics, including means and standard deviations, for all study variables. Additionally, Spearman’s correlation coefficients were computed to explore the relationships between the key constructs, providing preliminary insights into the associations among the variables. To test the proposed hypotheses, we employed structural equation modelling (SEM) to examine the relationships between the variables. Specifically, all hypotheses (H1–H11) were examined as hypothesised direct paths specified in the structural model. Model fit was evaluated using multiple fit indices: the Comparative Fit Index (CFI) and Tucker–Lewis Index (TLI) with values ≥ 0.90 indicating acceptable fit, and Root Mean Square Error of Approximation (RMSEA) and Standardised Root Mean Square Residual (SRMR) with values ≤ 0.08 considered indicative of good fit (Kline, 2023). In addition to testing the direct relationships specified in the structural model, indirect effects involving PE, PU, and emotion were examined using bootstrapping procedures (2000 resamples, 95% confidence interval) to explore potential mediation pathways, although these indirect effects were not specified as formal hypotheses. While the proposed model specifies directional pathways grounded in appraisal theory, the analyses are based on observational data; accordingly, all estimated paths and mediation effects should be interpreted as theory-driven associations rather than definitive causal relationships. All analyses were conducted using SPSS 26.0 and Mplus 8.3.

4. Results

4.1. Descriptive results

Table 1 shows the mean, standard deviation, and measurement weights for each item for the whole sample. The 27 items had skewness values in the range of 0.55 to 0.33 (below the recommended threshold value of 3.0) and kurtosis values in the range of -0.88 to 0.88 (below the recommended threshold value of 3.0). Given that skewness and kurtosis are well within the recommended thresholds, the assumption of approximate normality was considered tenable (Kline, 2023). The mean scores for individual items were in the range of 3.98–5.86, with SDs

Table 1

The means, standard deviations, and measurement weights (standardised factor loadings and p-value) for each item for the whole sample.

Variable	Item	Mean	SD	Factor loading	p
GenAI competency	I know how to interact with GenAI-based tools in daily life.	5.46	1.00	0.93	<0.001
	I know how to execute some tasks with GenAI-based tools.	5.47	0.96	0.83	<0.001
	I know how to initialise a task for GenAI-based technologies by text or speech.	5.46	1.06	0.89	<0.001
	I have sufficient knowledge to use GenAI-based tools.	5.16	1.13	0.81	<0.001
	I am familiar with GenAI-based tools and their technical capacities.	5.07	1.13	0.80	<0.001
Academic self-efficacy	Regarding today’s task, compared with other students in this class I expect to do well.	4.68	1.06	0.87	<0.001
	Regarding today’s task, I expect to do very well in this class.	4.58	1.06	0.91	<0.001
	Regarding today’s task, my study skills are excellent compared with others in this class.	4.44	1.08	0.88	<0.001
	Regarding today’s task, I am sure I can do an excellent job on the problems and tasks assigned for this class.	4.62	1.04	0.89	<0.001
Perceived ease of use	Regarding today’s task, my interaction with chatbot is clear and understandable.	5.45	1.04	0.83	<0.001
	Regarding today’s task, it’s easy for me to become skilled at using chatbot.	5.24	1.11	0.87	<0.001
	Regarding today’s task, I find chatbot easy to use.	5.27	1.14	0.83	<0.001
	Regarding today’s task, I think using chatbot improves my task performance.	5.69	0.85	0.86	<0.001
Perceived usefulness	Regarding today’s task, I think using chatbot increases my productivity.	5.86	0.85	0.90	<0.001
	Regarding today’s task, I think using chatbot enhances my effectiveness on the task.	5.76	0.91	0.89	<0.001
	Regarding today’s task, I think using chatbot makes it easier to do my task.	5.79	0.91	0.90	<0.001
	When I use chatbots to finish today’s task, I feel melancholic – contented.	4.03	0.73	0.83	<0.001
Emotion	When I use chatbot to finish today’s task, I feel bored – relaxed	4.01	0.75	0.83	<0.001
	When I use chatbot to finish today’s task, I	4.07	0.71	0.87	<0.001

(continued on next page)

Table 1 (continued)

Variable	Item	Mean	SD	Factor loading	p
Engagement in chatbot-supported learning	feel despairing – hopeful				
	When I use chatbot to finish today's task, I feel unsatisfied – satisfied	4.02	0.74	0.90	<0.001
	When I use chatbot to finish today's task, I feel annoyed – pleased	3.98	0.72	0.88	<0.001
	Regarding today's task, while using chatbot, I am satisfied with the quality of my learning experiences.	4.76	0.74	0.78	<0.001
	Regarding today's task, I become more interested in using chatbot for learning.	4.79	0.79	0.81	<0.001
	Regarding today's task, chatbot keeps my attention.	4.73	0.81	0.84	<0.001
	Regarding today's task, I am involved in the learning experiences using chatbot.	4.78	0.80	0.94	<0.001
	Regarding today's task, I actively use chatbot in learning.	4.85	0.80	0.90	<0.001
	Regarding today's task, I feel my learning intellectually stimulating while using chatbot.	4.79	0.86	0.81	<0.001

ranging from 0.71 to 1.14.

Table 2 shows the Cronbach's α values, means and standardised deviations of the study variables for the whole sample, and the correlations between each pair of the factors. Statistically significant and strong positive correlations between the six variables were observed ($r = 0.25\text{--}0.70$, $p < .001$).

The results showed that the single factor accounted for 49% of the total variance, which is below the commonly accepted threshold of 50% (Podsakoff et al., 2003). This suggests that CMV is unlikely to pose a serious concern in this study.

4.2. The measurement model and structural model

The measurement model demonstrated a good fit to the data (CFI = 0.952, TLI = 0.946, RMSEA = 0.059, SRMR = 0.042). All standardised

Table 2

The means and standard deviations of the study variables for the whole sample and the correlations between each pair of factors.

Variables	1	2	3	4	5	6
1. GenAI competency	-					
2. Academic self-efficacy	0.37**	-				
3. Perceived ease of use	0.68**	0.42**	-			
4. Perceived usefulness	0.56**	0.33**	0.66**	-		
5. Emotion	0.49**	0.25**	0.51**	0.57**	-	
6. Learning engagement	0.60**	0.25**	0.61**	0.60**	0.70**	-
M	5.32	4.58	5.32	5.77	4.02	4.78
SD	0.94	0.97	0.98	0.81	0.65	0.70
Cronbach's alpha	0.94	0.94	0.88	0.94	0.94	0.94

Note: Learning engagement means Engagement in chatbot-supported learning. ** $p < .001$ level (2-tailed).

factor loadings exceeded the recommended threshold of 0.50, indicating satisfactory factor structure (Hair et al., 2019). The regression analysis revealed that the participants' gender and age do not have a significant effect on the examined variables. Therefore, these two demographic factors were not controlled in subsequent analyses. SEM was used to test the hypothesised model and yielded a good model fit ($\chi^2 = 649.899$, $df = 357$, $\chi^2/df = 1.82$, $p < .0001$, CFI = 0.952, TLI = 0.946, RMSEA = 0.059, SRMR = 0.046) (see Fig. 2).

The findings indicate that GenAI competency was critical to students' PE ($\beta = 0.75$, $p < .001$), emotion ($\beta = 0.18$, $p < .05$), and learning engagement ($\beta = 0.37$, $p < .001$), and PU ($\beta = 0.22$, $p < .05$). Accordingly, H1–H4 were all supported. Academic self-efficacy was found insignificantly related to students' emotional responses and their learning engagement, thus neither H5 nor H6 is supported. PE significantly predicted PU ($\beta = 0.57$, $p < .001$), supporting H7. Notably, PE and PU, both demonstrated significant positive associations with students' emotional responses, thereby supporting H8 ($\beta = 0.24$, $p < .05$) and H9 ($\beta = 0.31$, $p < .001$). Because the relationship between PU and learning engagement did not reach statistical significance, H10 was not supported. Finally, emotion was found to positively predict learning engagement ($\beta = 0.53$, $p < .001$), thus providing support for H11. The model accounted for a substantial proportion (69%) of the variance of learning engagement.

4.3. Mediation analysis

The mediation analysis was conducted to assess the mediating roles of PE, PU, and emotion. The results (see Table 3) highlighted the crucial role of emotion as a mediator in multiple ways. First, PU was significantly associated with learning engagement through the mediation of emotion ($\beta = 0.16$, $p < .01$). Meanwhile, PE did not have a significant indirect effect on engagement through PU or through emotion alone. However, when both PU and emotion were included as mediators, PE showed a significant indirect effect on engagement ($\beta = 0.10$, $p = .01$).

Regarding the relationship between GenAI competency and learning engagement, the partially mediated pathway, where GenAI competency was associated with cognitive appraisal factors (PE, PU) and emotion, and was subsequently linked to learning engagement, was also significant ($\beta = 0.07$, $p = .01$).

5. Discussion

Grounded in appraisal theory, this study is among the first to investigate the antecedents of student engagement in chatbot-supported learning. The findings indicate that GenAI competency plays a dual role: it shows a strong direct association with engagement, while also functioning as an initial appraisal that shapes students' perceptions of chatbot usefulness and their positive emotional responses. In this process, positive emotion emerges as a critical mediator linking appraisal to behavioural engagement. Taken together, these results advance an integrated understanding of how cognitive appraisals and emotional mechanisms jointly shape student engagement in chatbot-supported learning.

5.1. GenAI competency as a dual driver of engagement

Consistent with prior research on online learning engagement across educational technologies such as learning management systems, intelligent tutoring systems, and immersive learning environments (Aldowah et al., 2025; Hu & Xiao, 2025), the present study demonstrates that GenAI competency is a significant predictor of students' engagement in chatbot-supported learning. However, this study extends existing literature by revealing a dual pathway through which GenAI competency operates. Specifically, beyond its direct association with learning engagement, GenAI competency is also indirectly linked to engagement through in-situ cognitive appraisals, including PE and PU, as well as

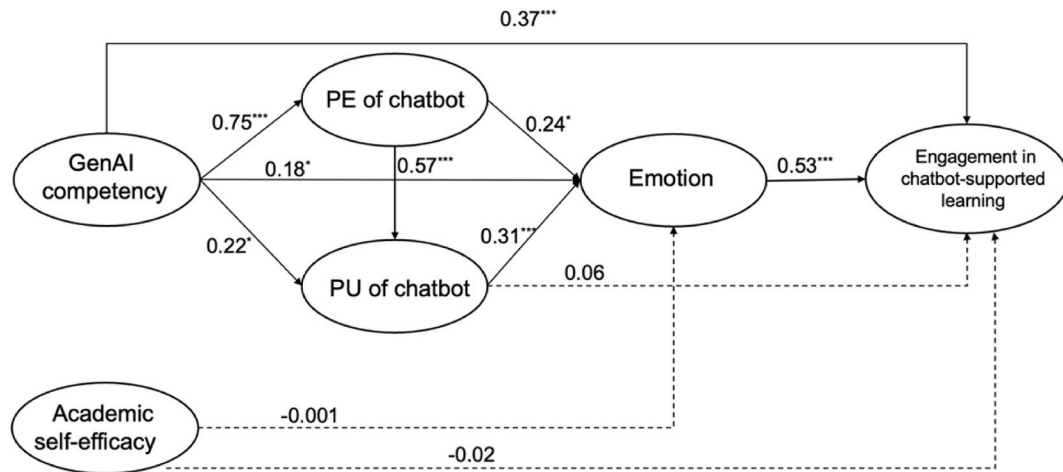


Fig. 2. Standardised regression coefficients of SEM.

Note: * $p < .05$; ** $p < .01$; *** $p < .001$. Chatbot means LLM-based chatbots.

Table 3

Standardised indirect effects of the hypothesised model.

Model Pathways	Point Estimate	p	95% CI	
			Lower 2.5%	Upper 2.5%
<i>From PU to Engage</i>				
PU→Emotion→Engage	0.16	0.005	0.056	0.283
<i>From PE to Engage</i>				
PE→PU→Engage	0.04	0.458	-0.049	0.140
PE→Emotion→Engage	0.12	0.099	-0.001	0.281
PE→PU→Emotion→Engage	0.10	0.010	0.035	0.186
<i>From AcaSE to Engage</i>				
AcaSE→Emotion→Engage	0.00	0.991	-0.064	0.063
<i>From GenAI competency to Engage</i>				
GenAI competency→PU→Engage	0.01	0.493	-0.013	0.070
GenAI competency→Emotion→Engage	0.10	0.078	-0.007	0.218
GenAI competency→PE→PU→Engage	0.03	0.457	-0.035	0.109
GenAI competency→PU→Emotion→Engage	0.04	0.134	0.002	0.100
GenAI competency→PE→Emotion→Engage	0.09	0.108	0.002	0.228
GenAI competency→PE→PU→Emotion→Engage	0.07	0.013	0.028	0.151

Note. PE means perceived ease of use, PU means perceived usefulness, AcaSE means academic self-efficacy, Engage means engagement in chatbot-supported learning.

affective appraisal reflected in learners' emotional responses. This pattern can be understood in light of the distinctive characteristics of LLM-based chatbots. As one of the most widely used GenAI tools in students' everyday learning activities, LLM-based chatbots require a relatively high level of user knowledge and skill to be employed effectively. Students with higher levels of GenAI competency are more likely to recognise both the affordances and limitations of chatbots (Tortella et al., 2025). They are also more experienced in formulating effective prompts, engaging in iterative dialogue with chatbots, and critically evaluating AI-generated content. Equipped with these competencies, students are less likely to encounter interactional or technical barriers and are therefore better able to concentrate on learning tasks themselves, which directly supports sustained engagement. Compared with more structured educational technologies, where technological skills are typically linked to engagement primarily through direct use-related mechanisms, the present findings suggest that in LLM-based learning contexts, the role of GenAI competency is more pronounced and operates through both direct and appraisal-based pathways.

In addition, the findings indicate that GenAI competency serves as an important antecedent of the positive emotions students experience during chatbot-supported learning. In line with existing GenAI-related studies (Subasi et al., 2024; Zhang et al., 2024), students with stronger GenAI competency are more likely to engage in smooth and productive interactions with chatbots, which can foster feelings of confidence, satisfaction, and enjoyment when learning processes or outcomes meet

or exceed expectations. These positive emotional experiences, in turn, are closely associated with higher levels of learning engagement, a mechanism that is examined in greater detail in the subsequent discussion.

5.2. The diminished role of academic self-efficacy

Surprisingly, academic self-efficacy, which has been widely recognised in previous studies as a significant predictor of engagement (Getenet et al., 2024; Luo et al., 2023; Zhen et al., 2017), did not emerge as significant in this study. This divergence suggests that in the context of chatbot-supported learning, students' general confidence in their academic abilities may be too distal to explain their actual engagement. Within chatbot-supported learning environments, engagement involves two interrelated demands: completing the assigned learning task and effectively communicating with the chatbot. The latter is complex because students need to adapt their prompts, interpret chatbot responses, and evaluate the relevance and accuracy of the generated content. Chatbot answers can be highly supportive, but they can also be distracting or confusing. When unexpected responses occur, as they frequently do, students must shift their attention away from the learning task to focus on managing the interaction, which makes it difficult to sustain concentration. Consistent with this view, prior research on on-line learning engagement has shown that factors related to learners' prior experience and interaction tend to exert a stronger influence on

engagement than academic skills (Aldowah et al., 2025). These findings suggest that GenAI competency constitutes a more proximal predictor of engagement than academic self-efficacy in chatbot-supported learning contexts.

The non-significant role of academic self-efficacy in predicting engagement can also be understood in light of the characteristics of the learning task. The focal activity required students to engage in instructional goal design, an open-ended and generative task with which most participants had little or no prior exposure, given their limited experience with authentic teaching or curriculum design. In such novel and ill-structured learning situations, learners may lack sufficient experiential grounding to form stable and task-relevant efficacy beliefs. As a result, general academic self-efficacy may remain tentative and less informative in explaining engagement during early encounters with unfamiliar tasks. This interpretation is consistent with prior research suggesting that the predictive role of self-efficacy is contingent upon task familiarity, structure, and the clarity of performance standards (Honicke et al., 2023).

5.3. The indirect effect of PE and PU on learning engagement

Contrary to findings from the broader educational technology literature (Al-Azawei et al., 2023; Hew et al., 2023), this study found that neither PE nor PU directly predicted student engagement in chatbot-supported learning. One possible explanation is that learning engagement reflects students' behavioural and cognitive involvement in the learning process (Halverson & Graham, 2019; Pentaraki & Burkholder, 2017), whereas PE and PU primarily reflect students' evaluations of the chatbot themselves. In other words, students' perceptions of whether the chatbot is easy to use or useful may not be sufficient to activate deeper engagement unless these perceptions are accompanied by other enabling conditions, such as emotional readiness. This highlights a potential limitation in directly applying traditional technology acceptance models to explain student engagement in pedagogically complex, chatbot-integrated environments.

5.4. The mediating role of emotion

This study highlights the pivotal role of emotion in shaping engagement in chatbot-supported learning. Beyond its direct association with engagement, emotion emerged as a key mechanism through which cognitive appraisals were translated into learners' behavioural involvement. Specifically, emotion fully mediated the relationship between PU and engagement and accounted for a substantial proportion of the total association between GenAI competency and engagement, indicating that emotional experience plays a proximal role in sustaining learners' engagement in chatbot-supported learning. This pattern differs from much of the prior technology acceptance and online learning research, in which emotion is typically positioned as a parallel antecedent alongside PE and PU (e.g., He et al., 2023; Jung & Lee, 2018; Nafees & Sujood, 2024). In contrast, the present findings suggest that in chatbot-supported learning contexts, emotion functions as a critical conduit through which learners' cognitive judgements are converted into sustained engagement. One plausible explanation lies in the distinctive nature of LLM interactions. Unlike more structured educational technologies, LLM-based chatbots involve open-ended, iterative, and often unpredictable exchanges, making learning progress closely dependent on the quality of ongoing interaction. In such contexts, learners' emotional responses may serve as immediate signals of interactional success or difficulty, thereby exerting a stronger influence on decisions to persist or disengage. The generative and exploratory nature of chatbot-supported tasks may further amplify emotional fluctuations during learning, elevating the role of emotion in mediating the relationship between cognitive appraisals and engagement. This interpretation is consistent with recent human-AI interaction research, which suggests that emotion in AI-mediated interaction is not merely a

background state but is actively elicited, intensified, and shaped through the interaction process itself (Gao et al., 2025; Riley et al., 2025). Taken together, these findings indicate that emotion occupies a more central and functionally distinct position in GenAI-supported learning than is typically assumed in prior educational technology research, underscoring the importance of explicitly accounting for affective mechanisms when examining engagement in chatbot-supported learning environments.

6. Implications

The findings of this study carry important implications for designing pedagogically meaningful environments that support student engagement in GenAI-supported learning. First, domain-specific technological competencies, such as GenAI competency, appear to exert greater influence on students' behavioural engagement than general academic self-beliefs. This highlights the need to reconceptualise learning support strategies. Rather than focusing solely on boosting students' academic confidence, educators should prioritise explicit training and scaffolding that strengthen learners' ability to interact effectively with new technologies such as LLM-based chatbots. Such preparation ensures that these tools serve as facilitators of learning rather than sources of distraction. Notably, participants in this study reported only moderately positive levels of GenAI competency ($M = 5.05$ out of 7), underscoring the necessity of equipping students with foundational GenAI knowledge and skills. By enhancing this competency, educators can directly improve learners' readiness, reduce barriers to meaningful interaction with chatbots, and ultimately foster greater engagement in technology-integrated learning environments.

Second, this study underscores the central mediating role of emotion in the learning process. Emotional experiences significantly shape the extent to which students engage with chatbots. This suggests that effective chatbot-supported environments should not only facilitate task completion but also deliberately cultivate positive emotions such as enjoyment, confidence, and curiosity. To achieve this, attention is needed both at the level of system design and at the level of classroom practice. From the design perspective, intuitive and user-friendly interfaces, adaptive feedback mechanisms, and structured prompt examples can help reduce anxiety and frustration. Such features reassure students when errors occur, encourage curiosity rather than confusion, and maintain focus on learning tasks. In parallel, teachers play a crucial role in shaping students' emotional experiences with chatbot. By modelling proper responses to unexpected or imperfect chatbot outputs, they help students reframe setbacks as opportunities for exploration rather than obstacles. Teachers can further scaffold emotional regulation by encouraging reflective pauses, offering strategies for managing confusion, and highlighting persistence and adaptivity as valuable skills in navigating GenAI tools. Taken together, these approaches highlight that cultivating positive emotions is not a peripheral concern but a central condition for sustaining motivation and deepening student engagement in chatbot-supported learning.

7. Limitation

Despite the study's robust theoretical foundation and methodological rigour, several limitations should be acknowledged. First, although the appraisal theory provided a valuable framework, the study primarily focused on individual-level factors influencing student learning engagement with GenAI technologies. Future research could extend this work by including contextual factors and GenAI features. Second, the study relied on self-reported quantitative measures, which may be subject to biases such as social desirability or inaccurate self-assessment. Future studies may consider incorporating multi-source observational data or adopting mixed-methods approaches to triangulate findings and enhance interpretive richness.

Third, the cross-sectional design of this research limited its ability to

capture the temporal dynamics of student engagement with GenAI. Given that attitudes, emotional responses, and technological proficiency may develop over time, longitudinal designs are needed to explore causal pathways and track the evolving impact of examined variables on student engagement in LLM chatbot-supported learning. In addition, although the present study analysed task-embedded survey responses as independent analytic units, future research that systematically collects task-level data nested within individuals would benefit from adopting multilevel analytic approaches to more precisely account for within- and between-person variance. Moreover, students in the present study were allowed to use different LLM-based chatbots (e.g. ChatGPT 4.0 and DeepSeek-R2) to support task completion. As the study did not aim to compare chatbot platforms, chatbot type was not included as an analytical variable. This may represent an additional source of variability, and future research could benefit from systematically examining how different chatbot interaction styles shape learners' emotional experiences and engagement.

Fourth, the participant sample was drawn from a specific educational and cultural context, which constrains the generalisability of the findings. The study involved master's students from predominantly non-technical backgrounds at a single university in Hong Kong, and the results should therefore be interpreted with caution when extending to other educational levels, disciplinary domains, or cultural settings. For example, undergraduate learners, students from STEM disciplines, or individuals with extensive professional experience may exhibit different engagement patterns as task familiarity, technological exposure, and learning autonomy vary. In addition, engagement was examined over a relatively short time frame, and the role of academic self-efficacy may evolve with prolonged chatbot use as learners accumulate experience and develop more stable efficacy beliefs. Future research across diverse learner groups, disciplines, and time spans is needed to further examine the robustness and boundary conditions of the present findings. Fifth, the emotion construct in this study captured general affective valence rather than discrete emotional states. As different positive emotions (e.g. curiosity, interest, satisfaction) may have distinct effects on engagement, future research could adopt more fine-grained emotion measures to further unpack these relationships.

8. Conclusion

Grounded in appraisal theory, this study demonstrates that GenAI competency, rather than traditional predictors such as academic self-efficacy, is the primary driver of engagement in chatbot-supported learning. Positive emotions emerged as the key mechanism through which GenAI competency, along with perceptions of usefulness and ease of use, translated into sustained involvement. These findings extend the learning engagement literature by foregrounding domain-specific technological competencies and affective processes. They also call for pedagogical designs that not only strengthen students' GenAI skills but also foster positive emotions, both of which are essential for sustaining engagement in chatbot-supported learning.

CRedit authorship contribution statement

Xianhan Huang: Writing – review & editing, Writing – original draft, Resources, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Shiyu Zhang:** Writing – original draft, Formal analysis.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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